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18.102

Functional Analysis 18.102 — Functional Analysis May 2026 **EXTREMELY HARD**

– MIT LEVEL

Time Limit: 3 hours

Total Marks: 100

1. (5 points) Prove the Hille-Yosida theorem: a linear operator A generates a strongly continuous contraction semigroup on a Banach space X iff A is closed, densely defined, and all $\lambda > 0$ are in the resolvent set with $\|R(\lambda, A)\| \leq 1/\lambda$.

2. (3 points) State and prove the spectral theorem for compact self-adjoint operators on a Hilbert space. Show that such an operator has a countable set of eigenvalues that accumulate only at zero, and the eigenvectors form an orthonormal basis.

3. (6 points) Prove the Schauder fixed-point theorem using Schauder estimates and the Arzela-Ascoli theorem: a compact operator $T : X \to X$ on a Banach space has a fixed point if it maps a closed convex set into itself.

4. (4 points) Prove the Nash embedding theorem for compact Riemannian manifolds: any compact C^∞ Riemannian manifold can be isometrically embedded into $\mathbb{R}^N$ for sufficiently large N.

5. (5 points) Prove the Atiyah-Singer index theorem for a simple case: compute the index of the Dirac operator on S^2 and verify it equals the integral of the $\widehat{A}$-genus.

6. (3 points) Prove the Gagliardo-Nirenberg-Sobolev inequality: for $1 \leq p < n$, there exists $C = C(n, p) > 0$ such that $\|u\|_{L^{p^*}(\mathbb{R}^n)} \leq C\|\nabla u\|_{L^p(\mathbb{R}^n)}$ for all $u \in C_c^1(\mathbb{R}^n)$, where $p^* = np/(n-p)$.

7. (4 points) Construct an exhaustion function (proper smooth function bounded below) on any smooth manifold and use it to prove that every smooth manifold admits a complete Riemannian metric.

8. (4 points) Prove the maximum principle for elliptic PDEs: if $Lu = -\Delta u + c(x)u \geq 0$ in Ω with $c \geq 0$, then $\max_{\overline{\Omega}} u \leq \max_{\partial\Omega} u^+$. Show how this implies uniqueness for the Dirichlet problem.

9. (2 points) Let M be a compact oriented n-manifold with boundary ∂M. State and prove Stokes' theorem for differential forms. Then apply it to prove the divergence theorem: $\int_M \operatorname{div} F \, dV = \int_{\partial M} F \cdot n \, dS$.

10. (4 points) Prove that there is no continuous linear functional $L : \ell^\infty \to \mathbb{R}$ that extends the limit functional on c (convergent sequences) and satisfies $\|L\| = 1$. (Hint: use the Hahn-Banach theorem and consider the Banach limit.)

11. (3 points) Prove that symplectic integrators preserve the symplectic structure $\omega = \sum dq^i \wedge dp_i$ of Hamiltonian systems. Show that the Verlet (Störmer) method is symplectic and has energy error bounded over exponentially long times for integrable systems.

12. (6 points) Prove the Sobolev embedding theorem: for $p > n$, $W^{1,p}(\mathbb{R}^n) \hookrightarrow C^{0,\alpha}(\mathbb{R}^n)$ with $\alpha = 1 - n/p$. In particular, show that if $u \in W^{1,p}(\mathbb{R}^n)$ with $p > n$, then u is (almost everywhere equal to) a Hölder continuous function.

13. (3 points) Prove the Rellich-Kondrachov compactness theorem: $W^{1,p}(\Omega) \hookrightarrow\!\!\!\rightarrow L^q(\Omega)$ compactly for $1 \leq q < p^*$ when Ω is bounded and $\partial\Omega$ is C^1.

14. (5 points) Construct a sequence of continuous functions $f_n : [0,1] \to \mathbb{R}$ that converges pointwise to $f(x) = 0$ but does NOT converge uniformly. Prove your construction works and explain why uniform convergence fails.

15. (5 points) Compute the Fourier transform of the distribution $p.v.(1/x)$ and prove that $\widehat{p.v.(1/x)}(\xi) = -i\pi \ \mathrm{sgn}(\xi)$. Use this to solve the Hilbert transform equation $H(u) = f$.

16. (3 points) Let (X, d) be a complete metric space and $T : X \to X$ a contraction mapping with contraction factor $k < 1$. Prove the Banach fixed-point theorem: T has a unique fixed point. Additionally, derive the error estimate $\|x_n - x^*\| \leq \frac{k^n}{1-k}\|x_1 - x_0\|$ where $x_{n+1} = T(x_n)$.

17. (3 points) Prove that the Fourier transform is an isometry on $L^2(\mathbb{R}^n)$ (Plancherel's theorem) and extends uniquely from $L^1 \cap L^2$ to a unitary operator on L^2.

18. (3 points) Prove the Lax-Milgram theorem and use it to show existence and uniqueness of weak solutions to the Dirichlet problem $-\Delta u = f$ in Ω, $u|_{\partial\Omega} = 0$ for $f \in L^2(\Omega)$.

19. (4 points) Prove the open mapping theorem: if $T : X \to Y$ is a bounded linear operator between Banach spaces and T is surjective, then T maps open sets to open sets. Use it to prove the inverse mapping theorem.

20. (4 points) Prove the Poincaré inequality: for $u \in W_0^{1,p}(\Omega)$ with $\Omega \subset \mathbb{R}^n$ bounded, $\|u\|_{L^p} \leq C\|\nabla u\|_{L^p}$. Show that the constant C depends on the diameter of Ω and give the optimal constant for $p = 2$ on an interval.

21. (6 points) State and prove the Implicit Function Theorem. Apply it to show that the equation $F(x, y) = y^5 + y + x = 0$ can be solved for $y = g(x)$ near $(0, 0)$, and compute $g'(0)$.

22. (3 points) Prove the Hopf-Rinow theorem: for a Riemannian manifold (M, g), metric completeness is equivalent to geodesic completeness (every geodesic extends for all time).

23. (7 points) Prove using the epsilon-delta definition that the function $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$ is differentiable at $x = 0$ but $f'(x)$ is not continuous at $x = 0$. Carefully justify all steps in your epsilon-delta arguments.

24. (2 points) Let $f_n : [0, 1] \to \mathbb{R}$ be a sequence of functions of bounded variation with $\mathrm{Var}(f_n) \leq C$ uniformly and $f_n \to f$ pointwise. Prove that f is of bounded variation and $\mathrm{Var}(f) \leq C$. Give a counterexample showing that uniform bounded variation is necessary.

25. (3 points) Prove the Brouwer fixed-point theorem using differential topology: any continuous map $f : \overline{B}^n \to \overline{B}^n$ has a fixed point. Use the fact that there is no smooth retraction of the ball onto its boundary.

Solutions

Solution 1:

($\Rightarrow$): If A generates $T(t)$, show $R(\lambda, A) = \int_0^\infty e^{-\lambda t} T(t) dt$, so $\|R(\lambda, A)\| \leq \int_0^\infty e^{-\lambda t} \|T(t)\| dt \leq 1/\lambda$. ($\Leftarrow$): Define $A_n = nAR(n, A)$ (Yosida approximants) which are bounded generators of $T_n(t) = e^{tA_n}$. Show $T_n(t)$ converges strongly to a semigroup $T(t)$ and its generator is A.

Solution 2:

Let $T : H \to H$ be compact self-adjoint. Show $\|T\| = \sup_{|x|=1} |\langle Tx, x\rangle|$ (achieved). Let $\lambda_1 = \pm\|T\|$ be the extremal eigenvalue with eigenvector e_1. Inductively apply the same argument to T restricted to $\{e_1, \ldots, e_{n-1}\}^\perp$. Compactness implies $\lambda_n \to 0$. The eigenvectors form an ONB by the spectral decomposition $Tx = \sum \lambda_n \langle x, e_n\rangle e_n$.

Solution 3:

Let $C \subset X$ be closed convex, $T(C) \subset C$, T compact. Approximate T by finite-rank operators T_n using Schauder basis. By Brouwer, each T_n has a fixed point x_n. By compactness of T, $\{T(x_n)\}$ has convergent subsequence $T(x_{n_k}) \to y$. Since $x_{n_k} = T_{n_k}(x_{n_k})$ and $T_{n_k} \to T$ uniformly, y is a fixed point of T.

Solution 4:

Sketch: Use the implicit function theorem to construct a short immersion $f_0 : M \to \mathbb{R}^N$. The metric g on M can be expressed as $f_0^* g_{\mathbb{R}^N} + h$. Use the Nash-Moser iteration to solve the system of PDEs $\partial_i f \cdot \partial_j f = g_{ij}$. The iteration converges by tame estimates on the linearized operator $L_f(h) = df \cdot \nabla h$.

Solution 5:

The Dirac operator D on S^2 acts on spinors. $\mathrm{ind}(D) = \dim\ker D - \dim\ker D^*$. By Lichnerowicz formula, $D^2 = \nabla^*\nabla + \frac{1}{4}R$. On S^2 with $R = 2$, $\ker D$ consists of the constant spinor. So $\mathrm{ind}(D) = 1$. The $\widehat{A}$-genus: $\int_{S^2} \widehat{A}(TS^2) = \int_{S^2} 1 - \frac{p_1}{24} + \cdots = 1$ since $p_1(TS^2) = 2\chi = 4$ but the integral gives 0 at this order. Indeed $\int_{S^2} \frac{R}{4\pi} = \chi/2 = 1$.

Solution 6:

For $p = 1$, use $|u(x)| \leq \int_{-\infty}^{x_i} |\partial_i u| dx_i$, multiply over $i = 1, \ldots, n$, integrate, apply Hölder repeatedly to get $\|u\|_{n/(n-1)} \leq \prod(\int |\partial_i u|)^{1/n} \leq \frac{1}{n}\sum\int|\partial_i u|$. For general p, apply the $p = 1$ case to $|u|^q$ with suitable q and use Hölder.

Solution 7:

Let $\{U_i\}$ be a countable cover by coordinate balls, $\{\rho_i\}$ a partition of unity. Define $f = \sum_i i\rho_i$. This is proper: $f^{-1}([-\infty, c])$ is compact (only finitely many $i \leq c$ contribute). For a Riemannian metric, start with any metric g_0, let $g = g_0 + df \otimes df$ on $f^{-1}((-\infty, \infty))$. The gradient of f has unit length, so any curve must escape to infinity in infinite length.

Solution 8:

Suppose u attains an interior maximum at x_0 with $u(x_0) = M > 0$ and $M > \max_{\partial\Omega} u^+$. At x_0, $\Delta u \leq 0$ (Hessian negative semidefinite), so $Lu = -\Delta u + cu \geq 0 + cM \geq 0$. Strictly, if $c > 0$ or $M > 0$, this gives $Lu > 0$ at the maximum, contradiction. For uniqueness of $\Delta u = f$ with $u|_{\partial\Omega} = g$, difference of two solutions satisfies $\Delta w = 0$,

$w|_{\partial\Omega} = 0$, so $w \equiv 0$.

Solution 9:

Stokes: $\int_M d\omega = \int_{\partial M} \omega$ for an $(n-1)$-form ω. Proof: Use partition of unity to reduce to $\mathbb{R}^n$ case, where it follows from the fundamental theorem of calculus and Fubini. For divergence theorem, let $\omega = \sum_i (-1)^{i-1} F_i \, dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$. Then $d\omega = (\sum_i \partial F_i/\partial x^i) dx^1 \wedge \cdots \wedge dx^n = \operatorname{div} F \, dV$. The boundary term gives $\int_{\partial M} F \cdot n \, dS$ by the Hodge star duality.

Solution 10:

The Hahn-Banach theorem guarantees the existence of such extensions (Banach limits), but they are not unique and not continuous in the sense of being representable by a sequence. The key issue: ℓ^∞/c_0 is non-separable and the quotient map is not injective. Banach limits exist but require the axiom of choice. They satisfy $\liminf x_n \leq L(x) \leq \limsup x_n$ and are shift-invariant.

Solution 11:

A one-step method Φ_h is symplectic if $(\Phi_h)^*\omega = \omega$. For Verlet: $q_{n+1/2} = q_n + \frac{h}{2} M^{-1} p_n$, $p_{n+1} = p_n - h\nabla V(q_{n+1/2})$, $q_{n+1} = q_{n+1/2} + \frac{h}{2} M^{-1} p_{n+1}$. Direct computation shows the Jacobian satisfies $J^T \Omega J = \Omega$. By the backward error analysis, the numerical solution is the exact flow of a nearby Hamiltonian, giving $O(e^{-c/h})$ energy conservation over exponentially long times.

Solution 12:

For $u \in C_c^\infty(\mathbb{R}^n)$, write $u(x) - u(y)$ as an integral of ∇u over a chain connecting x and y. Using the mean value integral representation and Hölder inequality, $|u(x) - u(y)| \leq C \|\nabla u\|_{L^p} |x - y|^{1-n/p}$. By density of smooth functions in $W^{1,p}$, the inequality extends, proving the embedding is continuous.

Solution 13:

Take a bounded sequence $\{u_n\}$ in $W^{1,p}$. By the Sobolev-Gagliardo-Nirenberg inequality, it's bounded in L^{p^*}. Use the Frechet-Kolmogorov criterion for compactness in L^q: show $\|u_n(\cdot + h) - u_n\|_{L^q} \to 0$ uniformly as $h \to 0$ using difference quotients and the $W^{1,p}$ bound. Extract a subsequence convergent in L^q by the Arzela-Ascoli type argument.

Solution 14:

Let $f_n(x) = \max(0, 1 - n|x - 1/n|)$ (triangular spikes at $1/n$). For any x, for $n > 1/x$, $f_n(x) = 0$, so $f_n \to 0$ pointwise. But $\|f_n\|_\infty = 1$ for all n, so $\sup_{x\in[0,1]} |f_n(x) - 0| = 1 \not\to 0$, hence convergence is not uniform. This shows pointwise convergence does not imply uniform convergence.

Solution 15:

$\langle \widehat{p.v.(1/x)}, \varphi \rangle = \langle p.v.(1/x), \widehat{\varphi} \rangle$. Write $p.v.(1/x) = \lim_{\varepsilon \to 0^+} \frac{1}{x} \chi_{|x|>\varepsilon}$. Its Fourier transform: $\int_{|x|>\varepsilon} \frac{e^{-2\pi i x \xi}}{x} dx = -i\pi \operatorname{sgn}(\xi)$ as $\varepsilon \to 0$. The Hilbert transform $H(u) = \frac{1}{\pi} p.v. \int \frac{u(t)}{x-t} dt$ has symbol $\widehat{H(u)}(\xi) = -i \operatorname{sgn}(\xi) \widehat{u}(\xi)$, so $H^{-1} = -H$.

Solution 16:

Pick any $x_0 \in X$, define $x_{n+1} = T(x_n)$. Then $d(x_{n+1}, x_n) \le k^n d(x_1, x_0)$. By triangle inequality, $d(x_{n+m}, x_n) \le \sum_{i=0}^{m-1} k^{n+i} d(x_1, x_0) \le \frac{k^n}{1-k} d(x_1, x_0)$. So $\{x_n\}$ is Cauchy, converges to x^*. By continuity of T, $T(x^*) = x^*$. Uniqueness: if $Tx = x$, $Ty = y$, then $d(x, y) = d(Tx, Ty) \le kd(x, y)$, so $d(x, y) = 0$. The error estimate follows from taking $m \to \infty$.

Solution 17:

For $f \in L^1 \cap L^2$, define $\widehat{f}(\xi) = \int f(x) e^{-2\pi i x \cdot \xi} dx$. Show $\int |\widehat{f}|^2 = \int |f|^2$ by proving $\int \widehat{f} \bar{g} = \int f \bar{g}$ for g Schwartz (using Gaussian approximation and Fubini). Extend by density of $L^1 \cap L^2$ in L^2. The inverse Fourier transform gives the adjoint, and $\mathcal{F}^* = \mathcal{F}^{-1}$.

Solution 18:

Lax-Milgram: Let $a : H \times H \to \mathbb{R}$ be coercive bilinear continuous on Hilbert H, $f \in H^*$. Then $\exists! u \in H$: $a(u, v) = f(v)$ for all $v \in H$. For the Dirichlet problem, $H = H_0^1(\Omega)$, $a(u, v) = \int_\Omega \nabla u \cdot \nabla v$, $f(v) = \int_\Omega fv$. Poincaré inequality gives coercivity, trace theorem gives boundary values.

Solution 19:

Use Baire category: $Y = \cup_n nT(B_X(0, 1))$ so some $nT(B_X(0, 1))$ has nonempty interior. Show $T(B_X(0, 1))$ contains a neighborhood of 0. Then by scaling, T maps open balls to open sets. The inverse mapping theorem follows: if T is bijective bounded linear, then T^{-1} is bounded.

Solution 20:

For $u \in C_c^\infty(\Omega)$, write $u(x) = \int_{-\infty}^{x_1} \partial_1 u(t, x'') dt$. By Hölder, $|u(x)|^p \le (\operatorname{diam} \Omega)^{p-1} \int |\partial_1 u|^p dt$. Integrate over Ω to get $\|u\|_p^p \le (\operatorname{diam} \Omega)^p \|\nabla u\|_p^p$. On $(0, \pi)$, the optimal constant for $p = 2$ is $C = 1/\pi$, attained by $u(x) = \sin(x)$.

Solution 21:

IFT: Let $F : \mathbb{R}^{m+n} \to \mathbb{R}^n$ be C^1, $F(a, b) = 0$, and $\partial F/\partial y(a, b)$ invertible. Then there exists $g : U \to \mathbb{R}^n$ C^1 near a with $F(x, g(x)) = 0$ and $Dg(x) = -(\partial F/\partial y)^{-1} \partial F/\partial x$. For $F(x, y) = y^5 + y + x$, $\partial F/\partial y = 5y^4 + 1$, at $(0, 0)$ this is $1 \ne 0$, so IFT applies. $g'(0) = -(\partial F/\partial y)^{-1} \partial F/\partial x = -1/1 = -1$.

Solution 22:

($\Rightarrow$): Assume metric completeness. For a geodesic γ with maximal interval (a, b), if $b < \infty$, take $t_n \to b$. Then $\{\gamma(t_n)\}$ is Cauchy (distance $\le C|t_n - t_m|$) so converges. Extend γ using the exponential map at the limit point. ($\Leftarrow$): Assume geodesic completeness. Take a Cauchy sequence p_n. Connect each p_n to p_0 by a minimizing geodesic. Bound on distances and completeness of initial data gives convergent subsequence.

Solution 23:

We prove $f'(0) = 0$ by showing $|(f(h) - f(0))/h - 0| = |h \sin(1/h)| \le |h|$. For any $\varepsilon > 0$, choose $\delta = \varepsilon$. Then $|h| < \delta$ implies $|h \sin(1/h)| < \varepsilon$, so $f'(0) = 0$. For $x \ne 0$, $f'(x) = 2x \sin(1/x) - \cos(1/x)$. This oscillates wildly near 0 since $\cos(1/x)$ has no limit. Hence f' is not continuous at 0.

Solution 24:

For any partition $0 = x_0 < \cdots < x_k = 1$, $|f(x_i) - f(x_{i-1})| = \lim_n |f_n(x_i) - f_n(x_{i-1})| \leq \liminf_n \mathrm{Var}(f_n) \leq C$. So $\mathrm{Var}(f) \leq C$. Counterexample without UB: $f_n(x) = nx$ on $[0, 1/n]$, $f_n(x) = 1$ elsewhere. $\mathrm{Var}(f_n) = 2n$, $f_n \to 1$ pointwise, but limit has variation 0, so inequality fails direction.

Solution 25:

Suppose f has no fixed point. Define $r(x) =$ intersection of ray from $f(x)$ through x with ∂B^n. This gives a smooth retraction (after smooth approximation). By Sard's theorem, regular values exist. For a regular value y, $r^{-1}(y)$ is a 1-manifold with boundary, one endpoint at y. The other endpoint would be in B^n, impossible since r maps interior to boundary. Contradiction.